



Volumes of Revolution · Mark Scheme

Core Pure 2 Chapter 4: Volumes of Revolution. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – About the y-axis

$$1 \quad V = \pi \int_0^4 x^2 dy = \pi \int_0^4 y dy \quad M1; = 8\pi \quad A1.$$

$$2 \quad x^2 = \frac{4}{y^2} \quad B1; V = \pi \int_1^4 4y^{-2} dy = \pi [-4y^{-1}]_1^4 \quad M1; = 3\pi \quad A1.$$

$$3 \quad x^2 = y^{\frac{2}{3}} \quad B1; V = \pi \int_1^8 y^{\frac{2}{3}} dy = \pi \left[\frac{3}{5} y^{\frac{5}{3}} \right]_1^8 \quad M1; = \frac{3\pi}{5} (32 - 1) = \frac{93\pi}{5} \quad A1.$$

$$4 \quad \text{Intersections at } (0,0) \text{ and } (2,4) \quad B1; \text{ for } 0 \leq y \leq 4 \text{ the region lies between } x = \frac{y}{2} \text{ and } x = \sqrt{y} \quad M1; V = \pi \int_0^4 \left(y - \frac{1}{4}y^2 \right) dy = \pi \left[\frac{1}{2}y^2 - \frac{1}{12}y^3 \right]_0^4 \quad M1; = \pi \left(8 - \frac{16}{3} \right) = \frac{8\pi}{3} \quad A1.$$

$$5(a) \quad \frac{y}{2} = \ln(x+1) \Rightarrow x = e^{\frac{y}{2}} - 1 \quad M1; V = \pi \int_0^2 x^2 dy = \pi \int_0^2 \left(e^{\frac{y}{2}} - 1 \right)^2 dy, \text{ with limits } y = 0 \text{ (at } O) \text{ and } y = 2 \quad A1.$$

$$5(b) \quad \left(e^{\frac{y}{2}} - 1 \right)^2 = e^y - 2e^{\frac{y}{2}} + 1 \quad M1; \pi \left[e^y - 4e^{\frac{y}{2}} + y \right]_0^2 \quad M1 \quad A1; = \pi \left[(e^2 - 4e + 2) - (1 - 4) \right] = \pi (e^2 - 4e + 5) \quad A1.$$

$$5(c) \quad \text{Volume} = 5^3 \times \pi (e^2 - 4e + 5) \quad M1; = 595.3 \text{ cm}^3 \quad A1; \text{ mass} = 2.5 \times 595.3 = 1488 \text{ g} = 1.49 \text{ kg} \quad A1.$$

Section B – Parametric curves

$$6 \quad V = \pi \int_0^2 y^2 \frac{dx}{dt} dt = \pi \int_0^2 t^4 \times 2 dt \quad M1; = 2\pi \times \frac{32}{5} = \frac{64\pi}{5} \quad A1.$$

$$7 \quad x = 1 \Rightarrow t = 1, x = 9 \Rightarrow t = 3 \quad B1; V = \pi \int_1^3 4t^2 \times 2t dt = 8\pi \int_1^3 t^3 dt \quad M1; = 2\pi(81 - 1) = 160\pi \quad A1.$$

$$8 \quad V = \pi \int x^2 \frac{dy}{dt} dt \text{ with } \frac{dy}{dt} = 3t^2 \quad M1; = \pi \int_0^1 t^4 \times 3t^2 dt = 3\pi \int_0^1 t^6 dt \quad A1; = \frac{3\pi}{7} \quad A1.$$

$$9 \quad t^2 + 1 = 2 \Rightarrow t = 1; t^2 + 1 = 5 \Rightarrow t = 2 \text{ (as } t \geq 0) \quad B1; \frac{dx}{dt} = 2t, \text{ so } V = \pi \int_1^2 4t^2 \times 2t dt \quad M1; = 8\pi \left[\frac{1}{4}t^4 \right]_1^2 \quad M1; = 2\pi(16 - 1) = 30\pi \quad A1.$$

$$10(a) \quad \frac{dx}{dt} = -3 \sin t \quad B1; x = 0 \text{ when } t = \frac{\pi}{2} \text{ and } x = 3 \text{ when } t = 0, \text{ so } V = \pi \int_{\frac{\pi}{2}}^0 4 \sin^2 t (-3 \sin t) dt \quad M1; = 12\pi \int_0^{\frac{\pi}{2}} \sin^3 t dt \quad A1.$$

$$10(b) \quad \sin^3 t = \sin t - \sin t \cos^2 t, \text{ so } \int_0^{\frac{\pi}{2}} \sin^3 t dt = [-\cos t + \frac{1}{3} \cos^3 t]_0^{\frac{\pi}{2}} = \frac{2}{3} \quad M1; V = 8\pi \quad A1.$$

$$10(c) \quad V = \pi \int x^2 \frac{dy}{dt} dt = \pi \int_0^{\frac{\pi}{2}} 9 \cos^2 t \times 2 \cos t dt \quad M1 \quad A1; = 18\pi \int_0^{\frac{\pi}{2}} (\cos t - \sin^2 t \cos t) dt = 18\pi \left[\sin t - \frac{1}{3} \sin^3 t \right]_0^{\frac{\pi}{2}} \quad M1; = 18\pi \times \frac{2}{3} = 12\pi \quad A1.$$

Roar-level challenge

$$11 \quad V = \pi \int_0^{2\pi} y^2 \frac{dx}{dt} dt \text{ with } \frac{dx}{dt} = 1 - \cos t \quad M1; V = \pi \int_0^{2\pi} (1 - \cos t)^3 dt \quad A1; (1 - \cos t)^3 = 1 - 3 \cos t + 3 \cos^2 t - \cos^3 t, \text{ with } \cos^2 t = \frac{1}{2}(1 + \cos 2t) \quad M1; \text{ over } [0, 2\pi] \text{ the integrals of } \cos t, \cos 2t \text{ and } \cos^3 t \text{ are zero, so } V = \pi \left(2\pi + \frac{3}{2} \times 2\pi \right) \quad M1; = 5\pi^2 \quad A1.$$